

FIRM3D: Fast ion reduced models in 3D

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Summary

The dynamics of energetic particle species, born from fusion reactions or plasma heating schemes, are critical for predicting the behavior of magnetic confinement fusion experiments and future fusion reactors. Given that energetic particles are largely collisionless, the orbits of Monte-Carlo samples drawn from a given distribution function can be efficiently integrated in given electromagnetic fields. In addition to the static magneto-hydrodynamic (MHD) equilibrium magnetic fields produced due to the electromagnetic coils in a fusion device, MHD waves are excited by and can transport energetic particle populations.

FIRM3D is a software suite for modeling of energetic particle dynamics in 3D magnetic fields. The core routines are based on SIMSOPT (Landreman et al., 2021), but have been extended to include additional physics and diagnostics that are not typically required in the optimization context. This standalone framework enables more modular development of FIRM3D with minimal dependencies.

Components of FIRM3D include:

- Interfaces with MHD equilibrium and wave stability software.
- CPU and GPU parallelized integration of the guiding center orbit equation.
- Orbit visualization and transport diagnostics, including Poincaré maps and weighted Birkhoff averaging.

Statement of need

Given recent advances by the stellarator optimization community (Landreman et al., 2022), stellarator equilibria have now been identified which satisfy many physics and engineering constraints for a fusion reactor. One of the critical features of a stellarator equilibrium is the ability to confine the guiding center trajectories of energetic particle species. Through concepts such as quasisymmetry, the presence of a hidden symmetry of the field strength which provides integrability of guiding center motion, the magnetic fields of stellarators can now be designed to have excellent energetic particle confinement.

However, there are likely to be perturbing electromagnetic fields that can transport energetic particles, such as magneto-hydrodynamic (MHD) waves. The class of MHD waves of primary concern for interaction with EP species are Alfvén eigenmodes (AEs). AEs are driven unstable by free energy in the EP distribution function, and they can resonantly transport EPs. Alfvénic activity is considered the major limitation to alpha confinement in a burning tokamak plasma (Gorelenkov et al., 2014). The interaction of Alfvén eigenmodes (AEs) with energetic particles has been shown to drive substantial flattening of the fast-ion profile in tokamak experiments (Heidbrink et al., 2008). Alfvénic activity has also been observed on several stellarator configurations, including HSX (Deng et al., 2009), CHS (Takechi et al., 2002), LHD (Toi et al., 2011), W7-AS (Weller et al., 1994), TJ-II (Melnikov et al., 2014),

W7-X (Rahbarnia et al., 2020), and Heliotron-J (Yamamoto et al., 2007). Given the recent growth of the private fusion industry, several start-up companies pursuing the stellarator path to fusion are interested in assessing the stability of EP-driven waves and their impact on EP transport. The development of FIRM3D is, therefore, timely.

FIRM3D grew out of the guiding center integration routines in SIMSOPT, but has been extended to include additional physics and diagnostics specifically needed for energetic particle studies. The standalone framework enables more focused development of energetic particle physics capabilities with minimal dependencies, making it accessible to the broader stellarator and plasma physics community. The FIRM3D routines and their SIMSOPT precursors have been used in published research, such as a survey of EP loss mechanisms (Paul et al., 2022), transport driven by AEs (Paul et al., 2023), and trapped EP resonances (Chambliss et al., 2025).

Structure and capabilities

Integration of the guiding center trajectories is performed given the magnetic field, particle initial conditions, and integrator specification. The equilibrium magnetic field is typically generated through an interface with BOOZ_XFORM (Landreman, 2021) through the Boozer-MagneticField class. Since BOOZ_XFORM computes the Fourier harmonics of the magnetic field on a uniform grid in the radial direction, interpolation is used to determine the magnetic field throughout the volume. The magnetic field corresponding to an MHD mode from AE3D (Spong et al., 2010) or FAR3D (Varela et al., 2024) can then be superimposed on the interpolated equilibrium field. Such MHD modes are saved on a radial grid containing the Fourier harmonics in Boozer coordinates. Several helper functions are provided to generate particle initial conditions from a known distribution function, or by preserving some known conserved quantity such as the energy or canonical momentum. Several integrators can be selected. An interface to the BOOST Runge-Kutta Dormand-Prince 5 method (Boost.Numeric.Odeint, 2025) is provided, with adaptive step size control and dense output capabilities. To prevent the adaptive step size from becoming too small, a custom Dormand-Prince 5 method has also been implemented with minimum step size control based on (Press, 2007). Finally, a symplectic integrator for non-canonical guiding-center orbits is implemented using the explicit-implicit Euler scheme of (Albert et al., 2020).

Since the performance bottlenecks for FIRM3D are field interpolation and trajectory integration, the Lagrange interpolating polynomials and trajectory integrators are implemented in C++. Interfaces with python are provided through pybind11 (Jakob et al., 2017). MPI parallelization over Fourier harmonics and OpenMP parallelization over the number of interpolant nodes are provided to accelerate the field interpolant setup. Given the independence of guiding center trajectories, Monte-Carlo samples can be trivially parallelized over CPUs or GPUs. For GPU parallelization, CUDA kernels are implemented for field interpolation and trajectory integration.

Given the trajectory data, several transport diagnostics can be computed, such as Poincaré plots, characteristic orbit frequencies, weighted Birkhoff averaging (Duignan & Meiss, 2021), and orbit classification (Paul et al., 2022). Examples of some of these capabilities will be highlighted below.

Conservation properties

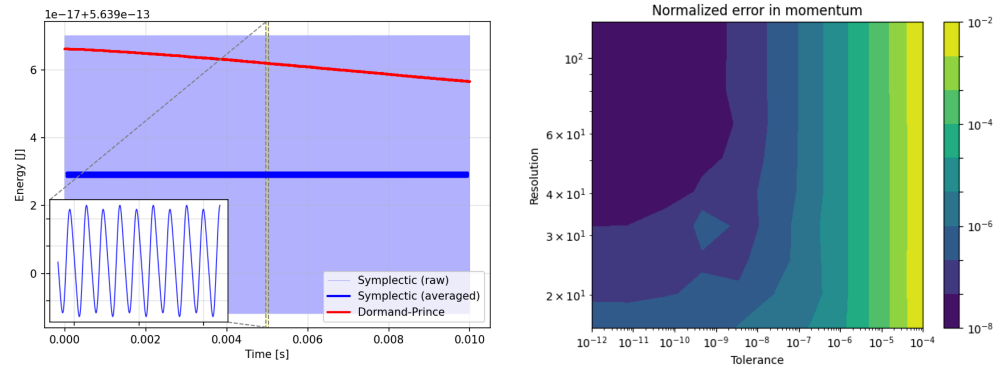


Figure 1: Left: Energy as a function of time for an alpha particle in the $\beta = 2.5\%$ Landreman QH configuration. The Dormand-Prince algorithm exhibits a net energy drift over time, while the symplectic algorithm exhibits a stable moving time average of the energy (over 10^{-4} seconds). Right: Relative error in canonical momentum P_η conservation for a perfectly quasisymmetric field. 10 particles are traced in the same configuration for 10^{-4} seconds, and the maximum error over the trajectory for each particle is computed. The maximum error over the 10 particles is reported. The non-quasisymmetric field-strength harmonics are artificially removed so that momentum conservation is expected.

Given a time-independent system, such as the guiding center Lagrangian in time-independent fields, the total energy E should be conserved. For a symplectic timestepper, while E is not precisely conserved throughout the trajectory, it demonstrates long-time stability, with a time-averaged value (over, e.g. several bounce periods) remaining conserved. On the other hand, Runge-Kutta methods suffer from net energy drift over time. This behavior is observed in Figure 1.

The accuracy of the integration methods depend on several resolution parameters, including the fixed timestep size and root solve tolerance for the symplectic method and the integration tolerance parameter for the Dormand-Prince method. Given the Lagrange field interpolation method, field smoothness will also impact the accuracy of integration. Thus improved conservation properties are observed at higher interpolant resolution. Given a perfectly quasisymmetric magnetic field, the toroidal canonical momentum P_η is also conserved (Boozer, 1995). In Figure 1 we show the relative error in momentum conservation as a function of the interpolant resolution parameter (the number of Lagrange interpolation nodes) and the tolerance parameter provided to the Dormand-Prince integrator. For an integration tolerance of 10^{-9} and grid resolution of 64, the relative error converges to around 10^{-8} .

Cross-code comparison

We perform a benchmark, Figure 2, with the SIMPLE code (Albert et al., 2020), which integrates the guiding center equations without MHD activity using a symplectic method. We first compare the trajectory of a trapped 3.5 MeV alpha particle in the precise QH equilibrium (Landreman & Paul, 2022). Given phase-space chaos, we don't generally expect good agreement between integrators for all trajectories, so the comparison is performed for a relatively integrable trajectory. With FIRM3D, the Dormand-Prince algorithm was used with a relative tolerance of 10^{-10} and 96 Lagrange interpolation nodes. With SIMPLE, the symplectic Euler method is used with 4096 timesteps per toroidal transit. The relative error in the s coordinate at 10^{-3} seconds is 7.8×10^{-3} . We next compare the loss fraction for a fusion birth distribution to demonstrate statistical agreement. 5000 3.5 MeV alpha particles are sampled from

a fusion birth distribution function and traced in the precise QA equilibrium (Landreman & Paul, 2022) for 10^{-2} seconds. The two codes report identical loss fractions at the end of the simulation. FIRM3D was run with 48 Lagrange interpolation nodes and a relative tolerance of 10^{-8} , and SIMPLE was run with 256 timesteps per toroidal transit.

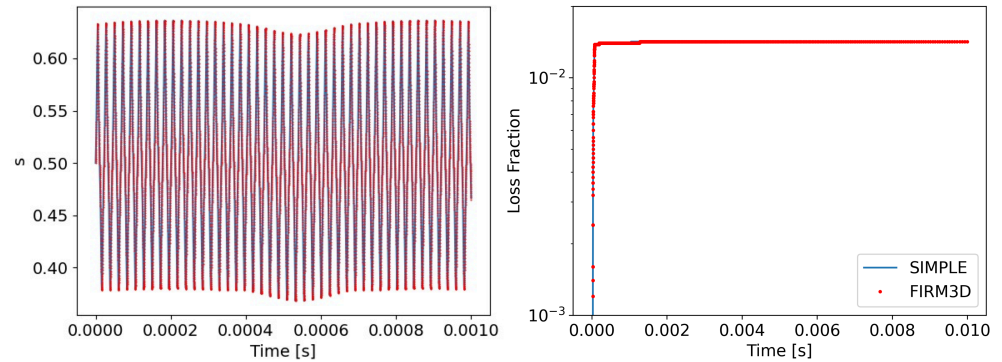


Figure 2: Left: Comparison of trapped 3.5 MeV alpha particle orbit in the precise QH equilibrium. The difference in the s coordinate between the trajectories at 10^{-3} seconds is 7.8×10^{-3} . Right: Comparison of loss fraction for the precise QA equilibrium. 5000 3.5 MeV alpha particles are sampled from a fusion birth distribution function and traced for 10^{-2} seconds. The two codes report identical loss fractions at the end of the simulation.

Scaling on GPUs and CPUs

In Figure 3 we show the scaling of wall clock time with CPU and GPU resources on the NERSC Perlmutter cluster. Particles are sampled from a fusion birth distribution in the Wistell-A equilibrium and integrated for 10^{-2} seconds. For a small number of Monte-Carlo samples, the CPU calculation is more efficient due to higher latency in the GPU calculation. However, for more than 10^3 samples, the GPU calculation is about an order of magnitude more efficient.

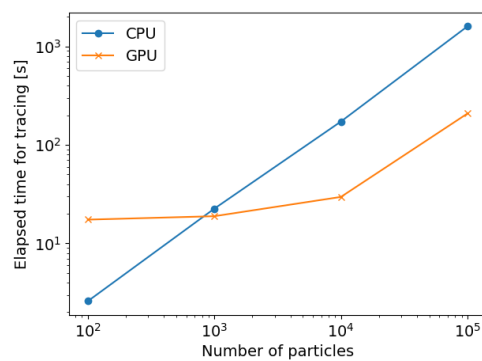


Figure 3: Scaling of tracing fusion birth distribution in Wistell-A equilibrium on 1 Perlmutter CPU node (128 CPUs) and 1 NVIDIA A100 with respect to number of Monte-Carlo samples.

Example applications

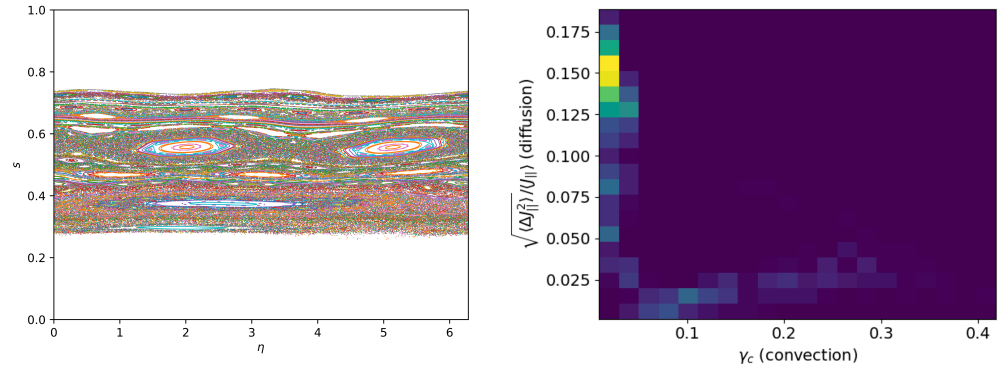


Figure 4: Left: Trapped particle Poincaré map showing chaotic layers responsible for banana-drift diffusion. Right: Measures of convective and diffusive transport indicate banana-trapped orbits undergo banana diffusion.

Given trajectory integration data, the transport characteristics can be identified. 5×10^5 particles are sampled from a fusion birth distribution function and traced in the $\beta = 2.5$ QA configuration (Landreman & Paul, 2022). 0.62% of samples are lost to the boundary in 10^{-2} seconds. The trapping class (banana, ripple, and barely trapped) can then be distinguished at any point in time along the trajectory. Of these losses, 0.12% are trapped in local ripple wells, 0.03% are barely trapped, 29% are in the banana class, and 4.8% transition between trapping classes. 66% of the lost samples exit the boundary promptly such that their trapping class cannot be identified. Given that banana-trapped particles represent the majority of losses with sufficient data to categorize, metrics of the diffusive and convective characteristics of the motion are shown in Figure 4. The parallel adiabatic invariant, $J_{\parallel} = \oint dl v_{\parallel}$, is an approximately conserved quantity for guiding center trajectories that lie on drift surfaces. The normalized variation in the parallel adiabatic invariant, $\sqrt{\langle \Delta J_{\parallel}^2 \rangle} / \langle J_{\parallel} \rangle$, is evaluated as a measure of integrability. A value of normalized variation of 1% is typically used as a cutoff for non-integrable orbits (Albert et al., 2023). The normalized variation of $J_{\parallel}$ is greater than 1% for most of the lost particles while γ_c , a measure of convective transport, is relatively small (< 0.2 (Paul et al., 2022)) over the population of lost particles. This data indicates that orbital chaos is responsible for the losses. The left panel of Figure 4 shows a Poincaré map revealing the chaotic layers responsible for banana-drift diffusion.

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References

- Albert, C. G., Buchholz, R., Kasilov, S. V., Kernbichler, W., & Rath, K. (2023). Alpha particle confinement metrics based on orbit classification in stellarators. *Journal of Plasma Physics*, 89(3), 955890301.
- Albert, C. G., Kasilov, S. V., & Kernbichler, W. (2020). Symplectic integration with non-canonical quadrature for guiding-center orbits in magnetic confinement devices. *Journal of Computational Physics*, 403, 109065.
- Boost.Numeric.Odeint. (2025). *Boost.odeint: Solving ordinary differential equations in C++*. <https://www.boost.org/libs/numeric/odeint>.
- Boozer, A. H. (1995). Quasi-helical symmetry in stellarators. *Plasma Physics and Controlled Fusion*, 37(11A), A103.
- Chambliss, A., Paul, E. J., & Hudson, S. R. (2025). Fast particle trajectories and integrability in quasisymmetric and quasihelical stellarators. *Journal of Plasma Physics*, 91(3), E74.
- Deng, C. B., Brower, D. L., Breizman, B. N., Spong, D. A., Almagri, A. F., Anderson, D. T., Anderson, F. S. B., Ding, W. X., Gutfenfelder, W., Likin, K. M., & others. (2009). Energetic-electron-driven instability in the Helically Symmetric Experiment. *Physical Review Letters*, 103(2), 025003.
- Duignan, N., & Meiss, J. D. (2021). Nonexistence of invariant tori transverse to foliations: An application of converse KAM theory. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 31(1).
- Gorelenkov, N. N., Pinches, S. D., & Toi, K. (2014). Energetic particle physics in fusion research in preparation for burning plasma experiments. *Nuclear Fusion*, 54(12), 125001.
- Heidbrink, W. W., Van Zeeland, M. A., Austin, M. E., Burrell, K. H., Gorelenkov, N. N., Kramer, G. J., Luo, Y., Makowski, M. A., McKee, G. R., Muscatello, C., & others. (2008). Central flattening of the fast-ion profile in reversed-shear DIII-D discharges. *Nuclear Fusion*, 48(8), 084001.
- Jakob, W., Rhineland, J., & Moldovan, D. (2017). *pybind11 – seamless operability between c++11 and python*.
- Landreman, M. (2021). https://github.com/hiddenSymmetries/booz_xform
- Landreman, M., Buller, S., & Drevlak, M. (2022). Optimization of quasi-symmetric stellarators with self-consistent bootstrap current and energetic particle confinement. *Physics of Plasmas*, 29(8).
- Landreman, M., Medasani, B., Wechsung, F., Giuliani, A., Jorge, R., & Zhu, C. (2021). SIMSOPT: A flexible framework for stellarator optimization. *Journal of Open Source Software*, 6(65), 3525.
- Landreman, M., & Paul, E. J. (2022). Magnetic fields with precise quasisymmetry for plasma confinement. *Physical Review Letters*, 128(3), 035001.
- Melnikov, A. V., Ochando, M., Ascasibar, E., Castejon, F., Cappa, A., Eliseev, L. G., Hidalgo, C., Krupnik, L. I., Lopez-Fraguas, A., Liniers, M., & others. (2014). Effect of magnetic configuration on frequency of NBI-driven Alfvén modes in TJ-II. *Nuclear Fusion*, 54(12), 123002.
- Paul, E. J., Bhattacharjee, A., Landreman, M., Alex, D., Velasco, J. L., & Nies, R. (2022). Energetic particle loss mechanisms in reactor-scale equilibria close to quasisymmetry. *Nuclear Fusion*, 62(12), 126054.
- Paul, E. J., Mynick, H. E., & Bhattacharjee, A. (2023). Fast-ion transport in quasisymmetric equilibria in the presence of a resonant alfvénic perturbation. *Journal of Plasma Physics*, 89(5), 905890515.

- Press, W. H. (2007). *Numerical recipes 3rd edition: The art of scientific computing*. Cambridge university press.
- Rahbarnia, K., Thomsen, H., Schilling, J., Vaz Mendes, S., Endler, M., Kleiber, R., Könies, A., Borchardt, M., Slaby, C., Bluhm, T., & others. (2020). Alfvénic fluctuations measured by in-vessel Mirnov coils at the Wendelstein 7-X stellarator. *Plasma Physics and Controlled Fusion*, 63(1), 015005.
- Spong, D. A., D'azevedo, E., & Todo, Y. (2010). Clustered frequency analysis of shear alfvén modes in stellarators. *Physics of Plasmas*, 17(2).
- Takechi, M., Matsunaga, G., Toi, K., Isobe, M., Minami, T., Tanaka, K., Nishimura, S., Takahashi, C., Okamura, S., & Matsuoka, K. (2002). Transition of toroidal Alfvén eigenmode to global Alfvén eigenmode in CHS heliotron/torsatron plasmas heated by neutral beam injection. *Journal of Plasma and Fusion Research*, 78(12), 1273–1274.
- Toi, K., Ogawa, K., Isobe, M., Osakabe, M., Spong, D. A., & Todo, Y. (2011). Energetic-ion-driven global instabilities in stellarator/helical plasmas and comparison with tokamak plasmas. *Plasma Physics and Controlled Fusion*, 53(2), 024008.
- Varela, J., Spong, D., Garcia, L., Ghai, Y., Ortiz, J., & Collaborators, F. P. (2024). Stability optimization of energetic particle driven modes in nuclear fusion devices: The FAR3d gyro-fluid code. *Frontiers in Physics*, 12, 1422411.
- Weller, A., Spong, D. A., Jaenicke, R., Lazaros, A., Penningsfeld, F. P., & Sattler, S. (1994). Neutral beam driven global Alfvén eigenmodes in the Wendelstein W7-AS stellarator. *Physical Review Letters*, 72(8), 1220.
- Yamamoto, S., Nagasaki, K., Suzuki, Y., Mizuuchi, T., Okada, H., Kobayashi, S., Blackwell, B., Kondo, K., Motojima, G., Nakajima, N., & others. (2007). Observation of magneto-hydrodynamic instabilities in Heliotron J plasmas. *Fusion Science and Technology*, 51(1), 92–96.